

# Companion to Bird & de Moor, Algebra of Programming

## §8. Thinning Algorithms

## §8.1. Thinning

(8.1)

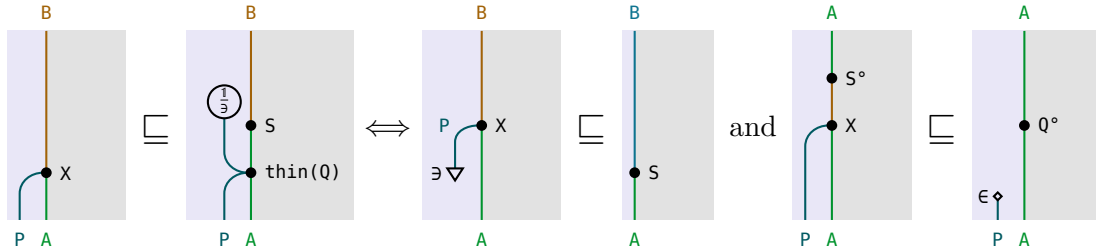
$\text{thin}(Q) \triangleq \sup_n (\epsilon \setminus (Q^\circ \epsilon))$  —  $\text{thin}(Q)$  takes a set  $ys$  to a subset  $xs$  that keeps a  $Q$ -lower bound of every  $a \in ys$ .  
 $\text{thin}(\leq) \ ys \ \text{thin}(\leq) \ xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \leq a).$

(8.1a)

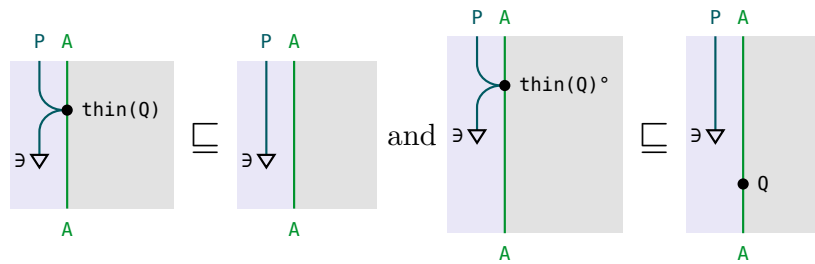
| the law                                                                                                                                                                                                             | what it says                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 $X \sqsubseteq \frac{S}{\sup} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \epsilon$                                                                                              | $S$ is the whole algebra: it produces every candidate that $\text{thin}(Q)$ thins. Everything kept is an $S$ -value, and every $S$ -value has a $Q$ -lower bound among the kept ones |
| 2 $\text{thin}(Q) \ni \sqsubseteq \ni$                                                                                                                                                                              | everything a thinning keeps was in the set                                                                                                                                           |
| 3 $\text{thin}(Q)^\circ \ni \sqsubseteq \ni Q$                                                                                                                                                                      | every element of the set has a $Q$ -lower bound among the kept ones                                                                                                                  |
| 4 $Q \sqsubseteq R \implies \text{thin}(Q) \sqsubseteq \text{thin}(R)$                                                                                                                                              | the fewer pairs $Q$ relates, the fewer subsets count as thinnings                                                                                                                    |
| 5 $1 \sqsubseteq Q \implies 1 \sqsubseteq \text{thin}(Q)$<br>$Q \sqsubseteq Q \implies \text{thin}(Q) \text{thin}(Q) \sqsubseteq \text{thin}(Q)$                                                                    | keeping everything is always a legal thinning                                                                                                                                        |
| 6 $Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge R R \sqsubseteq R \implies \text{est}(R) = \text{thin}(Q) \text{est}(R)$<br>$1 \sqsubseteq Q$<br>$\implies \text{est}(R) \sqsubseteq \text{thin}(Q) \text{est}(R)$ | <b>thin-introduction:</b> thinning first cannot lose an $R$ -minimum                                                                                                                 |
| 7 $\text{thin}(Q) \supseteq \text{est}(Q) \frac{1}{\sup} \quad (8.2) \text{ — } (8.2) \text{est}(Q) \frac{1}{\sup} \sqsubseteq \text{thin}(Q)$                                                                      | <b>thin-elimination:</b> keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on                                                                 |
| 8 $\frac{S}{\sup} \text{thin}(Q) \supseteq \frac{S}{\sup} \text{est}(R) \frac{1}{\sup}$<br>(8.3), $Rn(S^\circ S) \sqsubseteq Q$ — thin-elimination variant                                                          | the usable variant: $R$ need only refine $Q$ between values $S$ gives one argument                                                                                                   |
| 9 $P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q) \quad (8.4) \text{ — } (8.4)$<br>$P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q)$                                             | thinning each member set is a thinning of the union                                                                                                                                  |

§8.1.1.  $X \sqsubseteq \frac{S}{\sup} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \epsilon$  — the definition of  $\frac{S}{\sup} \text{thin}(Q)$ 

(8.1.1a)



(8.1.1b)



$X \sqsubseteq \frac{S}{\sup} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \epsilon$  at  $X \triangleq \text{thin}(Q)$ ,  $S \triangleq \ni$ : the left side is  $\text{thin}(Q) \sqsubseteq \text{thin}(Q)$ , since  $\frac{\ni}{\sup} = \wedge(\ni) = 1$

§8.1.2.  $Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge R R \sqsubseteq R \implies \text{est}(R) = \text{thin}(Q) \text{est}(R)$  — thinning first, then taking the smallest, is the same as taking the smallest

$$1 \sqsubseteq Q \Rightarrow \text{est}(R) \sqsubseteq \text{thin}(Q) \text{ est}(R) \quad (8.1.2a)$$

the  $\sqsubseteq$  half: keeping everything is a thinning —  $1 \sqsubseteq Q$



|   |                                             |                                                                                                                                                                                                                 |
|---|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $\text{est}(R)$                             | $1 \sqsubseteq \text{thin}(Q) \text{ — } 1 \sqsubseteq Q$<br>$\Rightarrow 1 \sqsubseteq \text{thin}(Q), Q$<br>$Q \sqsubseteq Q$<br>$\Rightarrow \text{thin}(Q) \text{ thin}(Q)$<br>$\sqsubseteq \text{thin}(Q)$ |
| 2 | $\sqsubseteq \text{thin}(Q) \text{ est}(R)$ |                                                                                                                                                                                                                 |

$$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists \quad (8.1.2b)$$

the  $\sqsubseteq$  half, first condition of the UP of  $\text{est}$  at  $X \triangleq \text{thin}(Q) \text{ est}(R) \text{ — } X \sqsubseteq \text{est}(R) \Leftrightarrow X \sqsubseteq \exists \wedge \in X \sqsubseteq R^\circ$



|   |                                      |                                              |
|---|--------------------------------------|----------------------------------------------|
| 1 | $\text{thin}(Q) \text{ est}(R)$      | $R \cap S \sqsubseteq R$                     |
| 2 | $\sqsubseteq \text{thin}(Q) \exists$ | $\text{thin}(Q) \exists \sqsubseteq \exists$ |
| 3 | $\sqsubseteq \exists$                |                                              |

$$Q \sqsubseteq R \wedge R R \sqsubseteq R \Rightarrow \in \text{thin}(Q) \text{ est}(R) \sqsubseteq R^\circ \quad (8.1.2c)$$

the  $\sqsubseteq$  half, second condition



|   |                                         |                                                                          |
|---|-----------------------------------------|--------------------------------------------------------------------------|
| 1 | $\in \text{thin}(Q) \text{ est}(R)$     | $\in \text{thin}(Q) \sqsubseteq Q^\circ \in$                             |
| 2 | $\sqsubseteq Q^\circ \in \text{est}(R)$ | $\in \text{est}(R) \sqsubseteq R^\circ$                                  |
| 3 | $\sqsubseteq Q^\circ R^\circ$           | $Q \sqsubseteq R$                                                        |
| 4 | $\sqsubseteq R^\circ R^\circ$           | $R R \sqsubseteq R$<br>$\Rightarrow R^\circ R^\circ \sqsubseteq R^\circ$ |
| 5 | $\sqsubseteq R^\circ$                   |                                                                          |

### §8.1.3. Properties of thinning

$$\mathbf{est}(Q) \stackrel{1}{\exists} \sqsubseteq \mathbf{thin}(Q)$$

(8.2)

the singleton holding a  $Q$ -least member of a set is a thinning of that set

$$\mathbf{est}(Q) \stackrel{1}{\exists} \ni \sqsubseteq \ni$$

every member of the singleton is a member of the set —  $\triangleq \mathbf{thin}, \ni/\ni$  half



|   |                                             |                                 |
|---|---------------------------------------------|---------------------------------|
| 1 | $\mathbf{est}(Q) \stackrel{1}{\exists} \ni$ | $\stackrel{1}{\exists} \ni = 1$ |
| 2 | $= \mathbf{est}(Q)$                         | $R \cap S \sqsubseteq R$        |
| 3 | $\sqsubseteq \ni$                           |                                 |

$$\in \mathbf{est}(Q) \stackrel{1}{\exists} \sqsubseteq Q^\circ \in$$

the singleton's member stands in  $Q$  to every member of the set —  $\triangleq \mathbf{thin}, \in/(Q^\circ \in)$  half

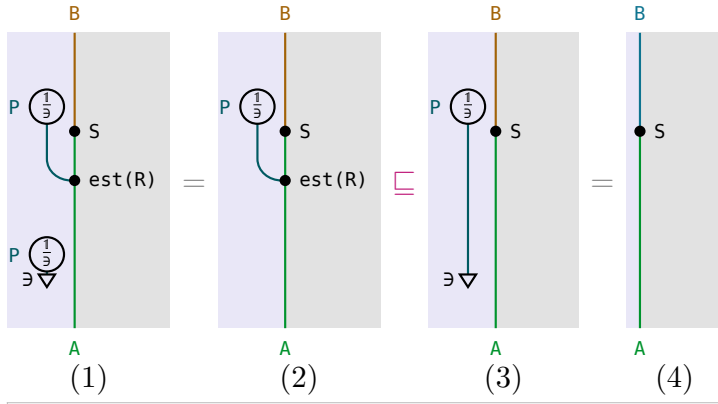


|   |                                             |                                           |
|---|---------------------------------------------|-------------------------------------------|
| 1 | $\in \mathbf{est}(Q) \stackrel{1}{\exists}$ | $\in \mathbf{est}(R) \sqsubseteq R^\circ$ |
| 2 | $\sqsubseteq Q^\circ \stackrel{1}{\exists}$ | $\stackrel{1}{\exists} \sqsubseteq \in$   |
| 3 | $\sqsubseteq Q^\circ \in$                   |                                           |

$$R \cap (S^\circ S) \subseteq Q \Rightarrow \frac{S}{S} \text{est}(R) \frac{1}{S} \subseteq \frac{S}{S} \text{thin}(Q) \quad (8.3)$$

$$\frac{S}{S} \text{est}(R) \frac{1}{S} \ni \in S$$

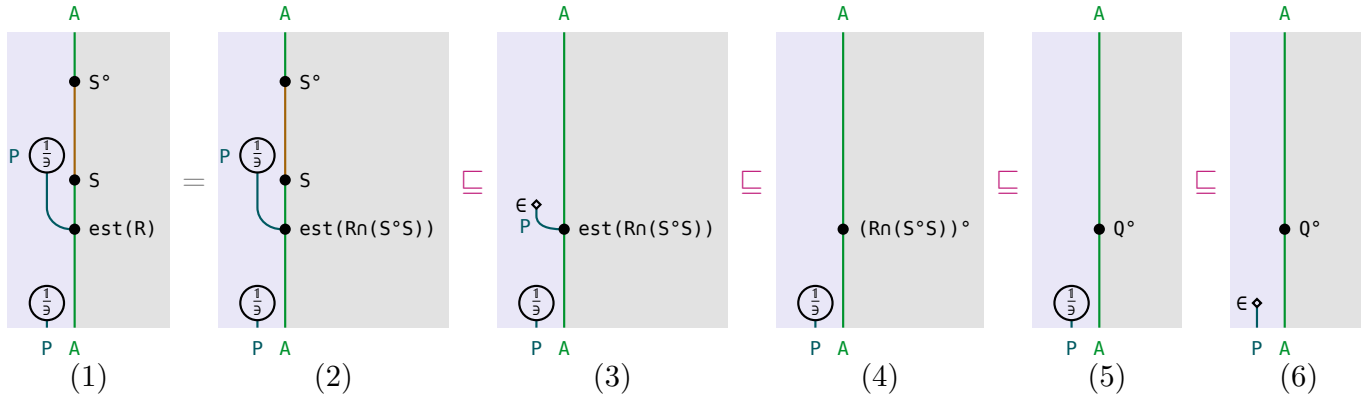
every member of the singleton is a value  $S$  returns —  $X \subseteq \frac{S}{S} \text{thin}(Q) \iff X \ni \in S \wedge S^\circ X \subseteq Q^\circ \in$ , first condition



|   |                                             |                        |
|---|---------------------------------------------|------------------------|
| 1 | $\frac{S}{S} \text{est}(R) \frac{1}{S} \ni$ | $\frac{1}{S} \ni = 1$  |
| 2 | $= \frac{S}{S} \text{est}(R)$               | $R \cap S \subseteq R$ |
| 3 | $\subseteq \frac{S}{S} \ni$                 | $\frac{R}{S} \ni = R$  |
| 4 | $= S$                                       |                        |

$$R \cap (S^\circ S) \subseteq Q \Rightarrow S^\circ \frac{S}{S} \text{est}(R) \frac{1}{S} \subseteq Q^\circ \in$$

the singleton's member stands in  $Q$  to every value  $S$  returns at the same argument —  $X \subseteq \frac{S}{S} \text{thin}(Q) \iff X \ni \in S \wedge S^\circ X \subseteq Q^\circ \in$ , second condition



|   |                                                                    |                                                                          |
|---|--------------------------------------------------------------------|--------------------------------------------------------------------------|
| 1 | $S^\circ \frac{S}{S} \text{est}(R) \frac{1}{S}$                    | $\frac{S}{S} \text{est}(R \cap (S^\circ S)) = \frac{S}{S} \text{est}(R)$ |
| 2 | $= S^\circ \frac{S}{S} \text{est}(R \cap (S^\circ S)) \frac{1}{S}$ | $T^\circ \frac{1}{S} \subseteq \in$                                      |
| 3 | $\subseteq \in \text{est}(R \cap (S^\circ S)) \frac{1}{S}$         | $\in \text{est}(R) \subseteq R^\circ$                                    |
| 4 | $\subseteq (R \cap (S^\circ S))^\circ \frac{1}{S}$                 | $R \cap (S^\circ S) \subseteq Q$                                         |
| 5 | $\subseteq Q^\circ \frac{1}{S}$                                    | $\frac{1}{S} \subseteq \in$                                              |
| 6 | $\subseteq Q^\circ \in$                                            |                                                                          |

$$\text{preorder}(Q) \wedge S \text{ Pres } Q \Rightarrow (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \sqsubseteq \frac{(S)}{\exists} \text{thin}(Q)$$

Theorem 8.1

thinning at every step of the reduce is a thinning of the whole candidate set

$$(\frac{F(\exists) S}{\exists} \text{thin}(Q)) \ni \sqsubseteq (S)$$

everything the thinning fold keeps is a value of the plain fold —  $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$ , first condition, by fusion from the chain below



|   |                                                     |                                          |
|---|-----------------------------------------------------|------------------------------------------|
| 1 | $\frac{F(\exists) S}{\exists} \text{thin}(Q) \ni$   |                                          |
| 2 | $= \frac{F(\exists) S}{\exists} \text{thin}(Q) \ni$ | $\text{thin}(Q) \ni \sqsubseteq \exists$ |
| 3 | $\sqsubseteq \frac{F(\exists) S}{\exists} \ni$      | $\frac{R}{\exists} \ni = R$              |
| 4 | $= F(\exists) S$                                    |                                          |

$$\text{preorder}(Q) \wedge S \text{ Pres } Q \Rightarrow (S)^\circ (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \sqsubseteq Q^\circ \in$$

every value of the plain fold is  $Q$ -above one the thinning fold keeps —  $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$ , second condition, by hylomorphism theorem from the chain below



|   |                                                                                       |                                                                         |
|---|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| 1 | $S^\circ F(Q^\circ \in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$                  |                                                                         |
| 2 | $= S^\circ F(Q^\circ) F(\epsilon) \frac{F(\exists) S}{\exists} \text{thin}(Q)$        | $S \text{ Pres } Q$                                                     |
| 3 | $\sqsubseteq Q^\circ S^\circ F(\epsilon) \frac{F(\exists) S}{\exists} \text{thin}(Q)$ | $\implies S^\circ F(Q^\circ) \sqsubseteq Q^\circ S^\circ$               |
| 4 | $\sqsubseteq Q^\circ \in \text{thin}(Q)$                                              | $S^\circ F(\epsilon) \frac{F(\exists) S}{\exists} \sqsubseteq \epsilon$ |
| 5 | $\sqsubseteq Q^\circ Q^\circ \in$                                                     | $\epsilon \text{thin}(Q) \sqsubseteq Q^\circ \in$                       |
| 6 | $\sqsubseteq Q^\circ \in$                                                             |                                                                         |

(8.1.3a)

| step                                                                                | type     | example                                        | what it does                                                                              |
|-------------------------------------------------------------------------------------|----------|------------------------------------------------|-------------------------------------------------------------------------------------------|
| S:F(A)→A is 8.1's algebra; example F(X)=V+V×X, S=[wrap,cons], candidate a=cons(v,p) |          |                                                |                                                                                           |
| S°                                                                                  | A→FA     | cons(v,p) ↦ r(v,p)                             | takes a apart into the input of its last step                                             |
| F(Q°∈)                                                                              | FA→F(PA) | r(v,p) ↦ r(v,ps), some p'∈ps with p' Q p       | replaces each recursive component by a set holding a Q-no-worse one; v stays              |
| $\frac{F(\exists)S}{\exists}$                                                       | F(PA)→PA | r(v,ps) ↦ {cons(v,t)   t∈ps}                   | picks one element from each set, combines by S, collects every result                     |
| thin(Q)                                                                             | PA→PA    | {cons(v,t)   t∈ps} ↦ ys                        | keeps a subset; every dropped element is Q-beaten by a kept one                           |
| $\sqsubseteq Q^\circ \in$                                                           | A→PA     | ys holds y with y Q cons(v,p') Q cons(v,p) = a | the thinned set still holds an element Q-no-worse than a (S monotonic on Q, Q transitive) |

$$Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge S \text{ Pres } Q \Rightarrow \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \text{est}(R) \sqsubseteq \frac{(S)}{\exists} \text{est}(R) \quad \text{Corollary 8.1}$$

the thinning fold refines the optimisation problem itself



$$\begin{aligned} 1 & \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \text{est}(R) \\ 2 & \sqsubseteq \frac{(S)}{\exists} \text{thin}(Q) \text{est}(R) \\ 3 & = \frac{(S)}{\exists} \text{est}(R) \end{aligned}$$

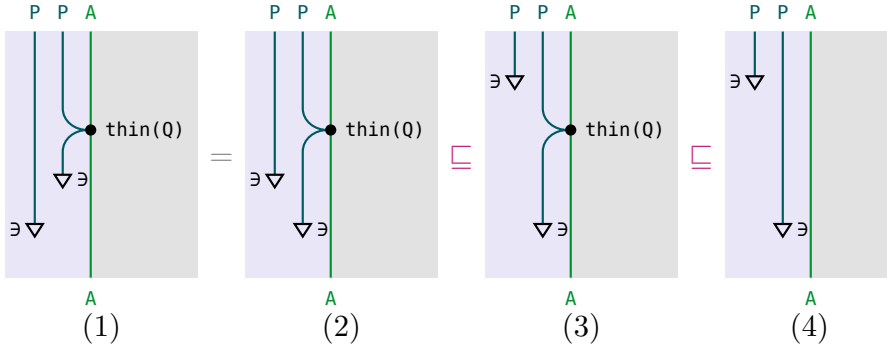
$$\begin{aligned} & \text{Theorem 8.1 } \text{preorder}(Q) \wedge S \text{ Pres } Q \\ & \Rightarrow \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \frac{(S)}{\exists} \text{thin}(Q) \end{aligned}$$

$$P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q) \quad (8.4)$$

thinning each member set, then taking the union, is a thinning of the union

$$P(\text{thin}(Q)) \text{ union} \exists \sqsubseteq \exists$$

every member of the result is a member of a member set —  $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \exists \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$ , first condition

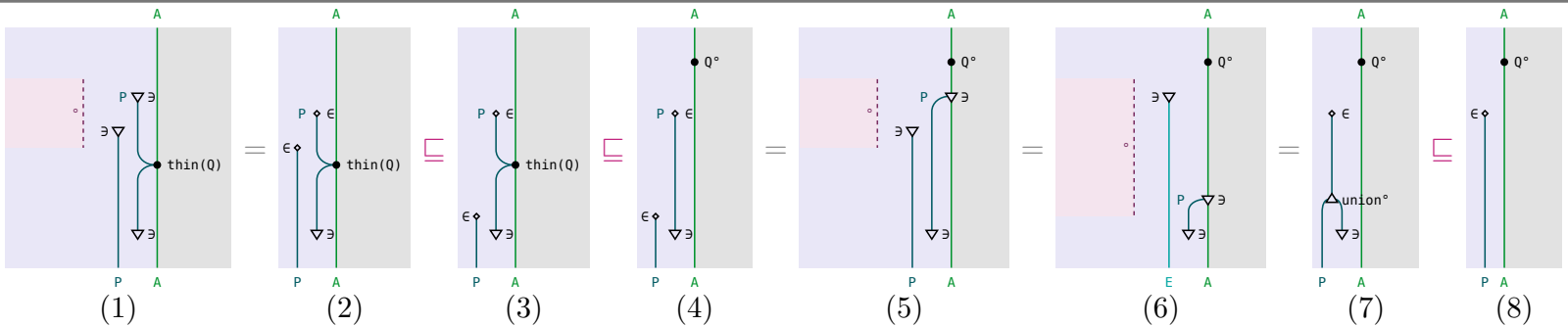


$$\begin{aligned} 1 & P(\text{thin}(Q)) \text{ union} \exists \\ 2 & = P(\text{thin}(Q)) \exists \\ 3 & \sqsubseteq \exists \text{thin}(Q) \exists \\ 4 & \sqsubseteq \exists \exists \end{aligned}$$

$$\begin{aligned} & \text{union } \exists = \exists \exists \\ & P(R) \exists \sqsubseteq \exists R \\ & \text{thin}(Q) \exists \sqsubseteq \exists \end{aligned}$$

$$(\exists \exists)^\circ P(\text{thin}(Q)) \text{ union} \sqsubseteq Q^\circ \in$$

every member of a member set has a  $Q$ -lower bound in the result —  $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \exists \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$ , second condition



$$\begin{aligned} 1 & (\exists \exists)^\circ P(\text{thin}(Q)) \text{ union} \\ 2 & = \in \in P(\text{thin}(Q)) \text{ union} \\ 3 & \sqsubseteq \in \text{thin}(Q) \in \text{union} \\ 4 & \sqsubseteq Q^\circ \in \in \text{union} \\ 5 & = Q^\circ (\exists \exists)^\circ \text{union} \\ 6 & = Q^\circ (\text{union } \exists)^\circ \text{union} \\ 7 & = Q^\circ \in \text{union}^\circ \text{union} \\ 8 & \sqsubseteq Q^\circ \in \end{aligned}$$

$$\begin{aligned} & (R S)^\circ = S^\circ R^\circ \\ & \in P(R) \sqsubseteq R \in \\ & \in \text{thin}(Q) \sqsubseteq Q^\circ \in \\ & (R S)^\circ = S^\circ R^\circ \\ & \text{union } \exists = \exists \exists \\ & (R S)^\circ = S^\circ R^\circ \\ & \text{union} = E(\exists) \end{aligned}$$

## §8.2. Paths in a layered network

(8.2a)

|    | definition                                                                   | type                                                               | meaning                                                                                                                                                                     |
|----|------------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | $F(B, X) = B + (B \times X)$                                                 | $\mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$           | a path is either one vertex $a$ , or a vertex $a$ followed by a path $xs$                                                                                                   |
| 2  | $\alpha = [\text{wrap}, \text{cons}]$                                        | $F(A, LA) \rightarrow LA$                                          | build a path: wrap one vertex as $[a]$ , or put $a$ in front of $xs$                                                                                                        |
| 3  | $\text{wrapz} \triangleq (\text{wrap}, \text{zero})$                         | $A \rightarrow LA \times \mathbb{N}$                               | a path of one vertex, no edge, cost 0                                                                                                                                       |
| 4  | $(a, (xs, n)) \mapsto ([a] \# xs, \text{wt}(a, \text{head}(xs)) + n)$        | $A \times (LA \times \mathbb{N}) \rightarrow LA \times \mathbb{N}$ | put $a$ in front of the path; the cost adds the weight of the new edge $a \rightarrow \text{head}(xs)$                                                                      |
| 5  | $\text{cost} \triangleq ([\text{wrapz}, \text{consw}]) \pi_2$                | $LA \rightarrow \mathbb{N}$                                        | the cost of $[a_0, \dots, a_n]$ is $\text{wt}(a_0, a_1) + \dots + \text{wt}(a_{n-1}, a_n)$                                                                                  |
| 6  | $R \triangleq \text{cost} \leq \text{cost}^\circ$                            | $LA \rightarrow LA$                                                | $xs$ is cheaper than $ys$                                                                                                                                                   |
| 7  | $([F(\exists, 1) \alpha]) = L(\exists)$                                      | $LA \rightarrow LB$                                                | $L(\exists)$ relates a list of layers to every path that chooses one vertex from each layer                                                                                 |
| 8  | $\text{minpath} \triangleq \frac{L(\exists)}{\exists} \text{est}(R)$         | $L(PA) \rightarrow LA$                                             | the input $[x_0, \dots, x_n]$ : $L(PA)$ is a list of layers, each a set of vertices; a path chooses one vertex from each layer, and $\text{minpath}$ returns a cheapest one |
| 9  | $Q \triangleq R \cap (\text{head} \text{ head}^\circ)$                       | $LA \rightarrow LA$                                                | like $R$ , and the two paths also start at the same vertex: $\text{head}(xs) = \text{head}(ys)$                                                                             |
| 10 | $a \mapsto l(a)$                                                             | $B \rightarrow B + C$                                              | mark a value as the left alternative                                                                                                                                        |
| 11 | $b \mapsto r(b)$                                                             | $C \rightarrow B + C$                                              | mark a value as the right alternative                                                                                                                                       |
| 12 | $\text{cpl} \triangleq \text{cp}(1 \times A)$                                | $PB \times A \rightarrow P(B \times A)$                            | pair every element of the set with the value beside it                                                                                                                      |
| 13 | $\text{cpr} \triangleq \text{cp}(A \times 1)$                                | $A \times PB \rightarrow P(A \times B)$                            | pair the value beside the set with every element of the set                                                                                                                 |
| 14 | $\frac{F(\exists, 1)}{\exists} = [P(l), \text{cpl } P(r)]$                   | $F(PB, X) \rightarrow PF(B, X)$                                    | $\text{cpl}$ transpose: from layer $x$ and a set of paths $ps$ , choose a vertex $a \in x$                                                                                  |
| 15 | $\frac{F(1, \exists)}{\exists} = [\frac{1}{\exists} P(l), \text{cpr } P(r)]$ | $F(B, PX) \rightarrow PF(B, X)$                                    | $\text{cpr}$ transpose: from a vertex $a$ and a set of paths $ps$ , choose a tail $xs \in ps$                                                                               |
| 16 | $\text{step} \triangleq \text{cpr } P(\text{cons}) \text{est}(R)$            | $A \times P(LA) \rightarrow LA$                                    | put $a$ in front of every path in $ps$ and keep a cheapest one                                                                                                              |

(8.2b)

| the law                                                                           | what it says                                                                                                                                                          |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 $F(\exists, Q) \alpha \sqsubseteq F(\exists, 1) \alpha Q$                       | $F(\exists, 1) \alpha$ is monotonic on $Q$ ; on $R$ it is not, since the next edge can cost arbitrarily much                                                          |
| 2 $S \text{ head} \sqsubseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$ | $S \text{ head}$ is simple, which gives $R n (S^\circ S) \sqsubseteq Q$ : between two paths $S$ builds from one argument, equal cost and equal head already means $Q$ |

## §8.2.1. Derivation

| $\frac{\frac{F(\exists, 1)}{\exists} \quad P(\frac{F(1, \exists) \alpha}{\exists} \text{ est}(R))}{\exists} \text{ est}(R) \sqsubseteq \frac{F(\exists, 1) \alpha}{\exists} \text{ est}(R)$ a least-cost path in a layered network, as a fold over the layers                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Why it is needed: <math>\text{est}(R)</math> keeps one cheapest <math>w</math> of the set <math>\Lambda(S)(x)</math>; <math>\text{thin}(Q)</math> needs every dropped <math>z</math> to be <math>Q</math>-beaten by a kept one. For <math>\{w\}</math> alone to be a thinning we need <math>w Q z</math> for every <math>z</math> in <math>\Lambda(S)(x)</math>.</li> <li>We have <math>w R z</math> (<math>w</math> is cheapest) and <math>w (S^\circ S) z</math> (both are built by <math>S</math> from the same argument <math>x</math>); the hypothesis turns the two into <math>w Q z</math>: among candidates from one argument, cheaper already means <math>Q</math>-better.</li> </ul> |                                                                                                                                                                     |
| <div style="display: flex; justify-content: space-around;"> <div>(1)</div> <div>(2)</div> <div>(3)</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                     |
| 1 $\frac{F(\exists, 1)}{\exists} \quad P(\frac{F(1, \exists) \alpha}{\exists} \text{ est}(R))}{\exists} \text{ est}(R)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\frac{F(\exists, 1)}{\exists} \quad P(\frac{F(1, \exists) \alpha}{\exists} \text{ est}(R)) \sqsubseteq \frac{F(\exists, \exists) \alpha}{\exists} \text{ thin}(Q)$ |
| 2 $\sqsubseteq \frac{F(\exists, \exists) \alpha}{\exists} \text{ thin}(Q) \text{ est}(R)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $\frac{F(\exists, \exists) \alpha}{\exists} \text{ thin}(Q) \text{ est}(R) \sqsubseteq \frac{F(\exists, 1) \alpha}{\exists} \text{ est}(R)$                         |
| 3 $\sqsubseteq \frac{F(\exists, 1) \alpha}{\exists} \text{ est}(R)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                     |

| $S \triangleq F(1, \exists) \alpha, R n (S^\circ S) \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} \quad P(\frac{F(1, \exists) \alpha}{\exists} \text{ est}(R)) \sqsubseteq \frac{F(\exists, \exists) \alpha}{\exists} \text{ thin}(Q)$ thinning the algebra of a layered network costs no more than taking the program's two cases                                                                                                                                 |                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-around;"> <div>(1)</div> <div>(2)</div> <div>(3)</div> <div>(4)</div> <div>(5)</div> <div>(6)</div> <div>(7)</div> </div>                                                                                                                                                                                                                                                                                                  |                                                                                                                |
| 1 $\frac{F(\exists, 1)}{\exists} \quad P(\frac{F(1, \exists) \alpha}{\exists} \text{ est}(R))$                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                |
| 2 $= \frac{F(\exists, 1)}{\exists} \quad P(\frac{F(1, \exists) \alpha}{\exists} \text{ est}(R))$                                                                                                                                                                                                                                                                                                                                                                             | $P(\frac{1}{\exists}) E(\exists) = 1$                                                                          |
| 3 $= \frac{F(\exists, 1)}{\exists} \quad P(\frac{F(1, \exists) \alpha}{\exists} \text{ est}(R) \frac{1}{\exists}) E(\exists)$                                                                                                                                                                                                                                                                                                                                                | (8.3) $R n (S^\circ S) \sqsubseteq Q$                                                                          |
| 4 $\sqsubseteq \frac{F(\exists, 1)}{\exists} \quad P(\frac{F(1, \exists) \alpha}{\exists}) P(\text{thin}(Q)) E(\exists)$                                                                                                                                                                                                                                                                                                                                                     | $\Rightarrow \frac{S}{\exists} \text{ est}(R) \frac{1}{\exists} \sqsubseteq \frac{S}{\exists} \text{ thin}(Q)$ |
| 5 $\sqsubseteq \frac{F(\exists, 1)}{\exists} \quad P(\frac{F(1, \exists) \alpha}{\exists}) E(\exists) \text{ thin}(Q)$                                                                                                                                                                                                                                                                                                                                                       | (8.4) $P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q)$                                       |
| 6 $= \frac{F(\exists, 1)}{\exists} \quad \frac{F(1, \exists) \alpha}{\exists} \text{ thin}(Q)$                                                                                                                                                                                                                                                                                                                                                                               | $\bigvee \frac{S}{\exists} = \bigvee \frac{P(\frac{S}{\exists})}{\exists} \text{ union}$                       |
| 7 $= \frac{F(\exists, \exists) \alpha}{\exists} \text{ thin}(Q)$                                                                                                                                                                                                                                                                                                                                                                                                             | $F(R, 1) F(1, S) = F(R, S)$                                                                                    |
| $= \text{cp}(F) P(\alpha) \text{ est}(R) = [\text{wrap}, \text{step}]$<br>at the layered network $(F(A, X) = A + A \times X, B \text{ the paths } V^+, \alpha = [\text{wrap}, \text{cons}])$ : $\frac{F(1, \exists)}{\exists} P(\alpha) \text{ est}(R) = [\text{wrap}, \text{step}] - \text{step} \triangleq \text{cpr}$<br>$P(\text{cons}) \text{ est}(R)$<br>$= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} - \frac{F(\exists, 1)}{\exists} = [P(l), \text{cpl } P(r)]$ |                                                                                                                |

$$\begin{array}{ccccc}
 F(A, X) & = & A & + & A \times X \\
 & & | & & | \quad | \\
 & & R & & R \quad S \\
 & & \downarrow & & \downarrow \quad \downarrow \\
 F(A', X') & = & A' & + & A' \times X'
 \end{array}$$

(8.2.1c)

u  $F(R, S) v$  holds iff one of:

- $u, v$  are both in  $l$  and  $d R d'$ : the leaf moves by  $R$ ;

- $u, v$  are both  $\text{inr}$  and  $p_1 \ R \ q_1 \wedge p_2 \ S \ q_2$ : the vertex moves by  $R$ , the rest by  $S$ .
- An  $\text{inl}$  never relates to an  $\text{inr}$ :  $R+S \models [R \ \text{inl}, S \ \text{inr}]$  returns to the summand it came from, and  $F(1)=1$  allows no crossing.
- $F(\exists, \exists)$  relates  $\text{inr}(\{a, b\}, \{p, q\})$  to  $\text{inr}(a, p), \text{inr}(a, q), \text{inr}(b, p), \text{inr}(b, q)$ , and  $\text{inl}(\{a, b\})$  to  $\text{inl}(a), \text{inl}(b)$ .

## §8.3. Implementing thin

(8.3a)

|    | definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | type                                                                        | meaning                                                                                                                                                                                                   |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | $(\text{setify } \exists)(xs, x) \iff x \in xs$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $[A] \rightarrow PA$                                                        | the set of a list's elements                                                                                                                                                                              |
| 2  | $\text{cup} \triangleq \frac{(\pi_1 \exists) \cup (\pi_2 \exists)}{\exists}$                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $PA \times PA \rightarrow PA$                                               | the union of two sets                                                                                                                                                                                     |
| 3  | $\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $F(PA) \rightarrow PFA$                                                     | every F-shaped value whose parts are chosen from the given sets                                                                                                                                           |
| 4  | $\text{listcpFn}(\text{zer}, x) = []$ , $\text{listcpFn}(\text{PolyC.one}, u) = [u, ()]$ ,<br>$\text{listcpFn}(\text{const}(a), c) = [c]$ , $\text{listcpFn}(\text{arg}_1, a) = [a]$ ,<br>$\text{listcpFn}(\text{arg}_2, xs) = xs$ , $\text{listcpFn}(l \otimes a, l(u)) =$<br>$\text{list}(\text{inl}, \text{listcpFn}(l, u))$ , $\text{listcpFn}(a \otimes r, r(v)) =$<br>$\text{list}(\text{inr}, \text{listcpFn}(r, v))$ , $\text{listcpFn}(l \otimes r, (u, v)) =$<br>$\text{cppFn}(\text{listcpFn}(l, u), \text{listcpFn}(r, v))$ | $\text{relator}(\text{list}(X)) \rightarrow \text{list}(\text{relator}(X))$ | by the shape of F: a constant or $\text{arg}_1$ is a one-element list, $\text{arg}_2$ the list itself, a sum lists its summand, a product pairs every element of one list with every element of the other |
| 5  | $\text{linear}(\text{zer}) = \text{true}$ , $\text{linear}(\text{PolyC.one}) = \text{true}$ , $\text{linear}(\text{const}(a)) =$<br>$\text{true}$ , $\text{linear}(\text{arg}_1) = \text{true}$ , $\text{linear}(\text{arg}_2) = \text{true}$ , $\text{linear}(l \otimes r) =$<br>$\text{linear}(l) \wedge \text{linear}(r)$ , $\text{linear}(l \otimes r) = \text{linear}(l) \wedge \text{linear}(r) \wedge (\neg$<br>$\text{hasArg}_2(l) \vee \neg \text{hasArg}_2(r))$                                                               | $P(\text{PolyC}(\iota))$                                                    | no product in F has the argument on both sides                                                                                                                                                            |
| 6  | $\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ordered}(\leq)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $PA \rightarrow LA$                                                         | read a set back as one of its $\leq$ -ordered listings                                                                                                                                                    |
| 7  | $(a, []) \mapsto [a]$ , $\text{bump}(Q)((a, [b] \# xs), ys) \iff Q(a, b) \wedge ys = [a] \# xs \vee$<br>$Q(b, a) \wedge ys = [b] \# xs \vee \neg Q(a, b) \wedge \neg Q(b, a) \wedge ys = [a] \# [b] \# xs$                                                                                                                                                                                                                                                                                                                              | $A \times [A] \rightarrow [A]$                                              | put a in front of a thinned list, dropping whichever of a and the old head the other beats                                                                                                                |
| 8  | $\text{thinlist}(Q) \triangleq (\text{setify thin}(Q) \text{ setify}^\circ) \cap \text{subseq}$                                                                                                                                                                                                                                                                                                                                                                                                                                         | $[A] \rightarrow [A]$                                                       | only drop elements, and on the set of the elements be a thinning by Q                                                                                                                                     |
| 9  | $T \sqsubseteq \text{thinlist}(Q) \iff T \sqsubseteq \text{subseq} \wedge T \text{ setify} \sqsubseteq \text{setify thin}(Q)$                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                             | what a list implementation of $\text{thin}(Q)$ must do: only drop elements, and thin the set                                                                                                              |
| 10 | $\text{preorder}(Q) \implies Q[\text{nil}, \text{bump}(Q)] \sqsubseteq \text{thinlist}(Q)$                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                             | an implementation: thin a list in one pass, bumping each element into the thinned rest                                                                                                                    |
| 11 | $\text{minlist}(Q) \triangleq \text{setify est}(Q)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $[A] \rightarrow A$                                                         | a Q-least element of the list                                                                                                                                                                             |

## §8.3.1. Sorting sets

(8.3.1a)

| the law                                                                                                                                                                                                        | what it says                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 1 $\text{preorder}(Q) \wedge \text{connected}(Q) \Rightarrow$<br>$\text{bumpFold}(Q)([a]\#xs,ys) \Leftarrow (\exists w. \text{minlist}(Q)([a]\#xs,w)$<br>$\wedge ys = [w])$ (8.5)                              | what thinning should come to when it can: one element                              |
| 2 $\text{sort}(\leq) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(\leq)$ (8.6) — (8.6)<br>$\text{sort}(\leq) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(\leq)$                     | the thinning may run after the sort                                                |
| 3 $\text{sort}(\leq) \text{minlist}(R) \sqsubseteq \text{est}(R)$ (8.7)                                                                                                                                        | a minimum of the sorted list is a minimum of the set                               |
| 4 $\text{sort}(f \leq f^\circ) \text{list}(f) \sqsubseteq P(f) \text{sort}(\leq)$ (8.8)                                                                                                                        | shunt a function through a sort                                                    |
| 5 $\text{Coreflexive}(p) \Rightarrow \text{sort}(\leq) \text{filter}(p) \sqsubseteq E(p) \text{sort}(\leq)$<br>(8.9)<br>$p : A \rightarrow A$                                                                  | filtering a sorted list sorts the restricted set                                   |
| 6 $\text{preorder}(\leq) \wedge \text{connected}(\leq) \Rightarrow (\text{sort}(\leq) \times \text{sort}(\leq))$<br>$\text{merge}(\leq) \sqsubseteq \text{cup sort}(\leq)$ (8.10)                              | merging two sorted lists sorts their union                                         |
| 7 $F(\text{sort}(\leq)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sort}(F(\leq))$<br>(8.11), $FX=L+E \times X$                                                                                              | $\text{listcp}$ is the list implementation of the cartesian product $\text{cp}(F)$ |
| 8 $\text{linear}(F) \Rightarrow \text{relator}(\text{sort}(\leq)) \text{listcp}(F,A,X) \sqsubseteq$<br>$\text{cp}(\text{relator}(F,A)) \text{sort}(\text{relator}(\leq))$<br>(8.11), $F$ polynomial and linear | the same for every linear polynomial $F$                                           |

 **$\text{sort}(\leq) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(\leq)$** 

(8.6)

every list got by sorting the set and then thinning it by  $Q$  is also got by thinning the set by  $Q$  and then sorting.

|                                                                       |                                                                                         |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| 1 $\text{sort}(\leq) \text{thinlist}(Q)$                              | $\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ordered}(\leq)$                 |
| 2 $\triangleq \text{setify}^\circ \text{ordered thinlist}(Q)$         | $\text{ordered thinlist}(Q) \sqsubseteq \text{thinlist}(Q) \text{ordered}$              |
| 3 $\sqsubseteq \text{setify}^\circ \text{thinlist}(Q) \text{ordered}$ | $\text{setify}^\circ \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{setify}^\circ$ |
| 4 $\sqsubseteq \text{thin}(Q) \text{setify}^\circ \text{ordered}$     |                                                                                         |
| 5 $= \text{thin}(Q) \text{sort}(\leq)$                                |                                                                                         |

| $\text{Coreflexive}(p) \wedge f \text{ Pres} \leq \Rightarrow F(\text{sort}(\leq)) \text{ listcp list}(f) \text{ filter}(p) \sqsubseteq \frac{F(\exists) f p}{\exists} \text{ sort}(\leq)$ <span style="float: right;">Lemma 8.1</span>                                                                                               |                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| sorting each argument set ( $F(\text{sort}(\leq))$ ), forming every combination of their elements ( $\text{listcp}$ ), applying $f$ to each ( $\text{list}(f)$ ) and keeping those that pass $p$ ( $\text{filter}(p)$ ) gives one of the sorted lists of the set of all $f(u)$ that pass $p$ , with $u$ drawn from the argument sets. |                                                                                       |
| <br>(1)                                                                                                                                                                                                                                                                                                                               | <br>(2)                                                                               |
| <br>(3)                                                                                                                                                                                                                                                                                                                               | <br>(4)                                                                               |
| <br>(5)                                                                                                                                                                                                                                                                                                                               | <br>(6)                                                                               |
| 1 $F(\text{sort}(\leq)) \text{ listcp list}(f) \text{ filter}(p)$                                                                                                                                                                                                                                                                     | $F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(F(\leq))$  |
| 2 $\sqsubseteq \text{cp}(F) \text{ sort}(F(\leq)) \text{ list}(f) \text{ filter}(p)$                                                                                                                                                                                                                                                  | $\text{Map}(f)$                                                                       |
| 3 $\sqsubseteq \text{cp}(F) \text{ sort}(f \leq f^\circ) \text{ list}(f) \text{ filter}(p)$                                                                                                                                                                                                                                           | $\Rightarrow f \text{ Pres } R \Leftrightarrow F(R) \sqsubseteq f R f^\circ$          |
| 4 $\sqsubseteq \text{cp}(F) P(f) \text{ sort}(\leq) \text{ filter}(p)$                                                                                                                                                                                                                                                                | $\text{sort}(f \leq f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(\leq)$     |
|                                                                                                                                                                                                                                                                                                                                       | $\text{Coreflexive}(p)$                                                               |
|                                                                                                                                                                                                                                                                                                                                       | $\Rightarrow \text{sort}(\leq) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(\leq)$ |
|                                                                                                                                                                                                                                                                                                                                       | $\text{Map}(f)$                                                                       |
|                                                                                                                                                                                                                                                                                                                                       | $\Rightarrow P(f) = E(f)$                                                             |
| 5 $\sqsubseteq \text{cp}(F) E(f p) \text{ sort}(\leq)$                                                                                                                                                                                                                                                                                | $\frac{S}{\exists} E(R) = \frac{S R}{\exists}$                                        |
| 6 $= \frac{F(\exists) f p}{\exists} \text{ sort}(\leq)$                                                                                                                                                                                                                                                                               |                                                                                       |

(8.3.1b)

At  $F(X)=L+Ex$ ,  $f(r(e,x))=e+x$ ,  $\leq$  is  $\leq$ ,  $p$  passes the odd numbers, input  $r(3,\{4,1,2\})$ .

| left side                | what it does                                     | result                        |
|--------------------------|--------------------------------------------------|-------------------------------|
| 1 input                  | an $r(e, xs)$ , $xs$ a set                       | $r(3, \{4, 1, 2\})$           |
| 2 $F(\text{sort}(\leq))$ | sorts the set in the $X$ place; $e$ is untouched | $r(3, [1, 2, 4])$             |
| 3 $\text{listcp}$        | pairs $e$ with each element, in list order       | $[r(3, 1), r(3, 2), r(3, 4)]$ |
| 4 $\text{list}(f)$       | $f$ on each                                      | $[4, 5, 7]$                   |
| 5 $\text{filter}(p)$     | keeps the odd ones, order kept                   | $[5, 7]$                      |

| right side            | what it does                                         | result                      |
|-----------------------|------------------------------------------------------|-----------------------------|
| 1 input               |                                                      | $r(3, \{4, 1, 2\})$         |
| 2 $F(\exists)$        | relates $r(3, xs)$ to each $r(3, x)$ with $x \in xs$ | $r(3, 4), r(3, 1), r(3, 2)$ |
| 3 $f$                 |                                                      | 7, 4, 5                     |
| 4 $p$                 | only odd values pass                                 | 7, 5                        |
| 5 $\wedge(\dots)$     | collects every result into one set                   | $\{5, 7\}$                  |
| 6 $\text{sort}(\leq)$ |                                                      | $[5, 7]$                    |

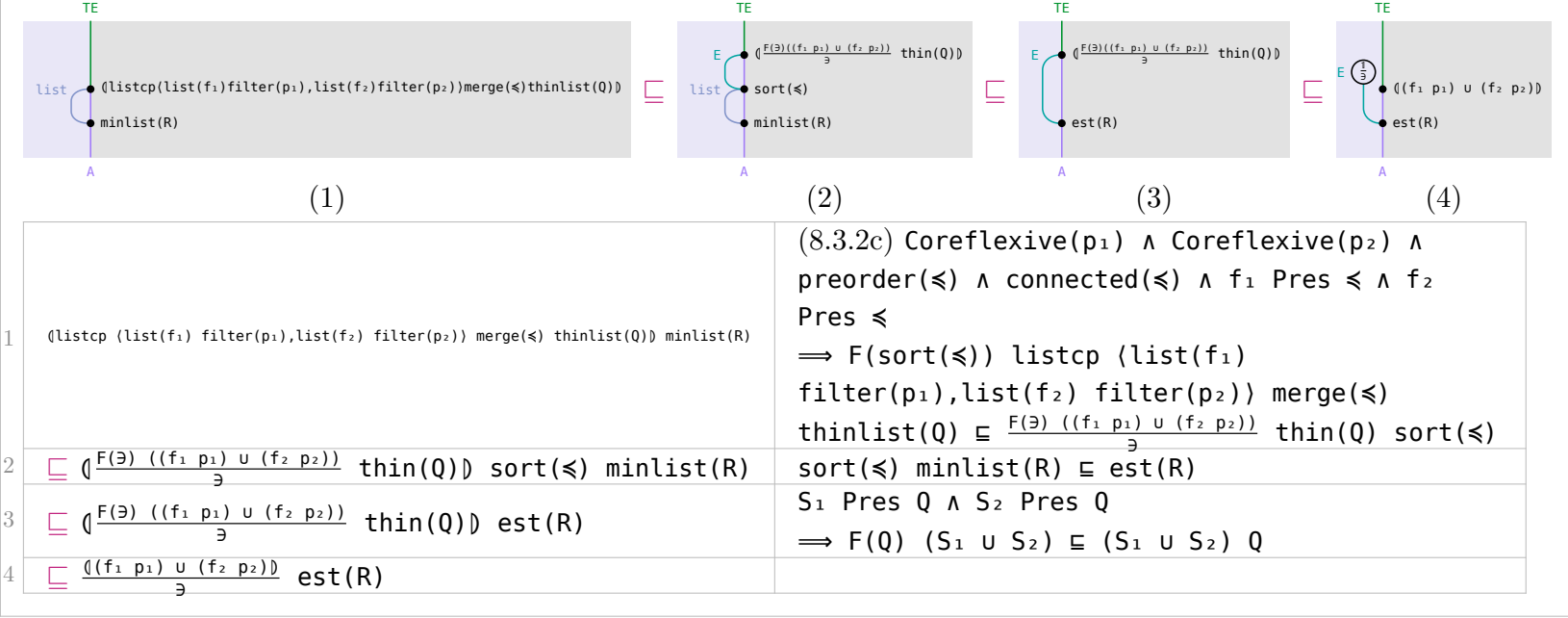
- $\sqsubseteq$ : the left's  $[5, 7]$  is one result of the right; with ties  $\text{sort}(\leq)$  has several orders.
- $f$  preserving  $\leq$  is used at  $\text{list}(f)$ :  $[1, 2, 4]$  is sorted, so  $[4, 5, 7]$  is;  $\text{filter}(p)$  only drops elements. With  $f(r(e, x))=e-x$  the left gives  $[1, -1]$  after  $p$ , not sorted, so  $\sqsubseteq$  fails.

### §8.3.2. Binary thinning

(8.3.2a)

**Binary thinning data:**  $Q$  a preorder with  $Q \sqsubseteq R$  and both  $f_1 p_1, f_2 p_2$  monotonic on  $Q$ ;  $\leq$  a connected preorder with both  $f_1, f_2$  monotonic on  $\leq$ ;  $R$  a preorder.

**Theorem 8.2**  
 $\text{Coreflexive}(p_1) \wedge \text{Coreflexive}(p_2) \wedge Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge (f_1 \text{ Pres } \leq \wedge f_2 \text{ Pres } \leq) \implies$   
 $Q \wedge (f_2 \text{ Pres } \leq) \wedge \text{preorder}(\leq) \wedge \text{connected}(\leq) \wedge f_1 \text{ Pres } \leq \wedge f_2 \text{ Pres } \leq \implies$   
 $(\text{listcp}(\text{list}(f_1) \text{ filter}(p_1), \text{list}(f_2) \text{ filter}(p_2)) \text{ merge}(\leq) \text{ thinlist}(Q)) \text{ minlist}(R) \sqsubseteq$   
 $\frac{Q((f_1 \text{ Pres } \leq) \cup (f_2 \text{ Pres } \leq))}{\exists} \text{ est}(R)$   
 a fold on sorted lists of partial solutions, thinned at every step, refines the thinning specification — at binary thinning data.

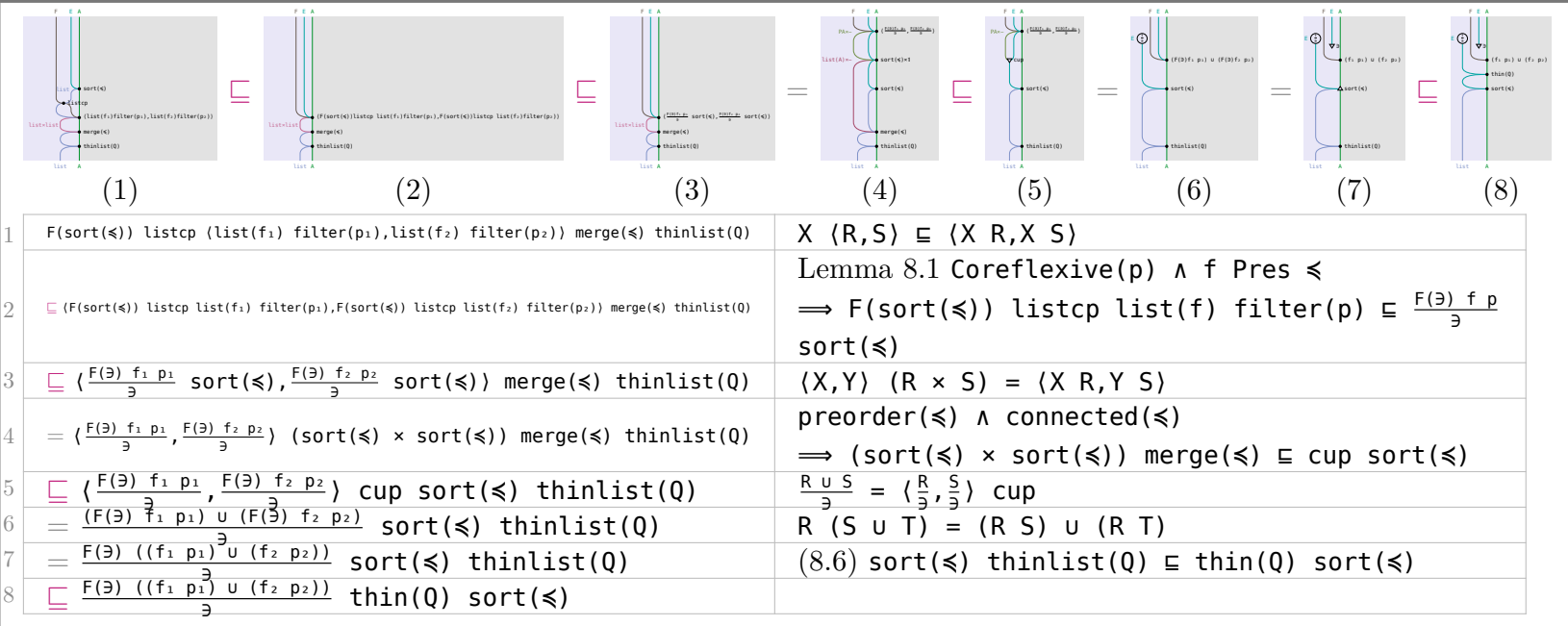


At  $F(X)=L+E \times X$ ,  $f_1(\text{inl}(l))=f_2(\text{inl}(l))=0$ ,  $f_1(r(e,x))=x$ ,  $f_2(r(e,x))=e+x$ ,  $p_1$  passes all,  $p_2$  passes values  $\leq 6$ ,  $\leq$  is  $\leq$ ,  $Q$  is  $=$ ,  $R$  is  $\geq$ , input  $[3,3,1]$ . (8.3.2b)

| input                 | $\text{list}(f_1) \text{ filter}(p_1)$ | $\text{list}(f_2) \text{ filter}(p_2)$ | $\text{merge}(\leq)$ | $\text{thinlist}(Q)$ |
|-----------------------|----------------------------------------|----------------------------------------|----------------------|----------------------|
| 1 $\text{inl}(l)$     | $[0]$                                  | $[0]$                                  | $[0,0]$              | $[0]$                |
| 2 $r(1,[0])$          | $[0]$                                  | $[1]$                                  | $[0,1]$              | $[0,1]$              |
| 3 $r(3,[0,1])$        | $[0,1]$                                | $[3,4]$                                | $[0,1,3,4]$          | $[0,1,3,4]$          |
| 4 $r(3,[0,1,3,4])$    | $[0,1,3,4]$                            | $[3,4,6]$                              | $[0,1,3,3,4,4,6]$    | $[0,1,3,4,6]$        |
| 5 $\text{minlist}(R)$ |                                        |                                        |                      | 6                    |

- $\text{merge}(\leq)$  is one pass: both copies arrive  $\leq$ -sorted ( $f_i$  monotonic on  $\leq$ ,  $\text{filter}(p_i)$  only drops), so it only compares their two heads.
- $\text{thinlist}(Q)$  compares neighbours only; the list is  $\leq$ -sorted, so equal values are neighbours and one pass drops every repeat.

**Theorem 8.3**  
 $\text{Coreflexive}(p_1) \wedge \text{Coreflexive}(p_2) \wedge \text{preorder}(\leq) \wedge \text{connected}(\leq) \wedge f_1 \text{ Pres } \leq \wedge f_2 \text{ Pres } \leq \implies$   
 $F(\text{sort}(\leq)) \text{ listcp}(\text{list}(f_1) \text{ filter}(p_1), \text{list}(f_2) \text{ filter}(p_2)) \text{ merge}(\leq) \text{ thinlist}(Q) \sqsubseteq$   
 $\frac{F(\exists) ((f_1 \text{ Pres } \leq) \cup (f_2 \text{ Pres } \leq))}{\exists} \text{ thin}(Q) \text{ sort}(\leq)$   
 sorting the candidate set turns the thinning algebra into an algebra on sorted lists.



## §8.4. The knapsack problem

(8.4a)

|   | definition                                                                                                                                                                                       | type                                     | meaning                                                     |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------|
| 1 | <code>total f = list(f) sum</code>                                                                                                                                                               | <code>Int</code>                         | add up f over the items of the list                         |
| 2 | <code>subseq = Q[<code>nil</code>,<code>cons</code> u <math>\pi_2</math>]]</code>                                                                                                                | <code>[A]→[A]</code>                     | every subsequence of the list                               |
| 3 | <code>[<code>nil</code>,<code>cons</code>] = [<code>nil</code>,<code>cons</code>]</code>                                                                                                         | <code>F[Item]→[Item]</code>              | keep the item: the list constructor                         |
| 4 | <code>[<code>nil</code>,<math>\pi_2</math>] = [<code>nil</code>,<math>\pi_2</math>]</code>                                                                                                       | <code>Unit + Item × [Item]→[Item]</code> | drop the item: keep the tail                                |
| 5 | <code>within(w)(xs,ys) ⇔ xs = ys ∧ total(wt,xs) ≤ w</code>                                                                                                                                       | <code>[Item]→[Item]</code>               | the packings whose total weight is at most w                |
| 6 | <code>([<code>nil</code>,<code>cons</code>] within(w)) u [<code>nil</code>,<math>\pi_2</math>] ≜ ([<code>nil</code>,<code>cons</code>] within(w)) u [<code>nil</code>,<math>\pi_2</math>]</code> | <code>F[Item]→[Item]</code>              | at each item, keep it if the packing still fits, or drop it |
| 7 | <code>R = total val ≥ total val°</code>                                                                                                                                                          | <code>[Item]→[Item]</code>               | x is worth at least as much as y                            |
| 8 | <code>Q(val,wt) = R n (total wt ≤ total wt°)</code>                                                                                                                                              | <code>[Item]→[Item]</code>               | x is worth at least as much as y and weighs no more         |

(8.4b)

|   | the law                                                                                                                                                                                                                                                    | what it says                                                                                   |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 1 | <code>(1×R) (cons (within w)) ⊆ cons (within w)R</code><br><code>FALSE</code>                                                                                                                                                                              | not monotonic on R: a selection of greater value need not still fit once one more item goes in |
| 2 | <code>F(Q(val,wt)) [<code>nil</code>,<code>cons</code>] within(w) ⊆ [<code>nil</code>,<code>cons</code>] within(w) Q(val,wt)</code><br><code>F(Q(val,wt)) [<code>nil</code>,<math>\pi_2</math>] ⊆ [<code>nil</code>,<math>\pi_2</math>] Q(val,wt)</code> , | both halves are monotonic on Q once ties in value are broken by weight                         |
| 3 | <code>F(R) [<code>nil</code>,<code>cons</code>] ⊆ [<code>nil</code>,<code>cons</code>] R</code><br><code>F(R) [<code>nil</code>,<math>\pi_2</math>] ⊆ [<code>nil</code>,<math>\pi_2</math>] R</code>                                                       | both algebras are monotonic on R itself, so R is the sort order P                              |

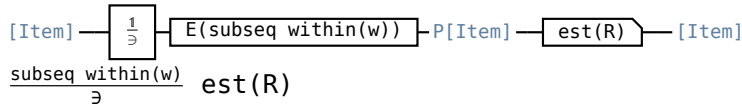
## §8.4.1. Derivation

$$0 \leq w \wedge (\forall i. 0 \leq \text{wt}(i)) \Rightarrow \langle \text{listcp } \langle g_1, g_2 \rangle \text{ merge}(R) \text{ thinlist}(Q(\text{val}, \text{wt})) \rangle \text{ minlist}(R) \sqsubseteq \text{est}(R) \quad (8.4.1a)$$

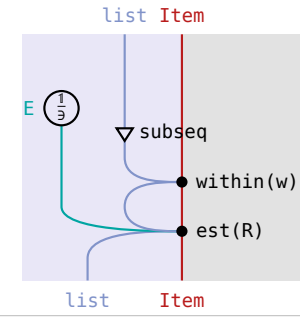
$\frac{\text{subseq within}(w)}{\exists}$

the knapsack problem, as a fold that thins the packings kept at each item

**circuit**



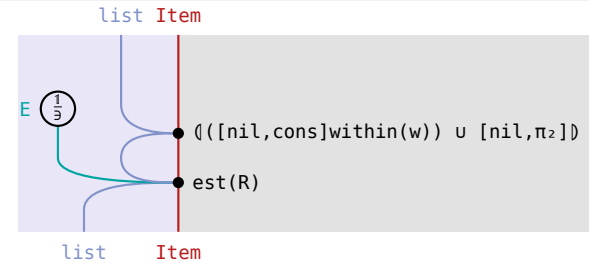
**Hinze–Marsden**



$$= \langle \langle [\text{nil}, \text{cons}] \text{ within}(w) \rangle \cup [\text{nil}, \pi_2] \rangle \text{ est}(R)$$

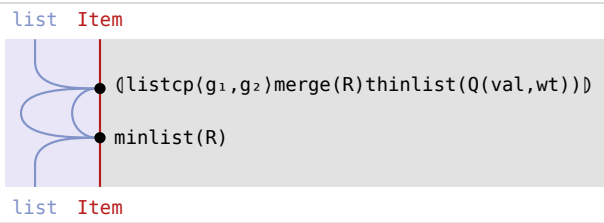
$\frac{\text{subseq within}(w)}{\exists}$

fusion, weights non-negative.



$$\langle \text{listcp } \langle g_1, g_2 \rangle \text{ merge}(R) \text{ thinlist}(Q(\text{val}, \text{wt})) \rangle \text{ minlist}(R)$$

binary thinning theorem, at  $P \triangleq R$ ,  $F$  linear,  $Q$  from  $F(Q(\text{val}, \text{wt}))$   
 $[\text{nil}, \text{cons}] \text{ within}(w) \sqsubseteq [\text{nil}, \text{cons}] \text{ within}(w) \ Q(\text{val}, \text{wt})$ ,  
 $F(Q(\text{val}, \text{wt})) [\text{nil}, \pi_2] \sqsubseteq [\text{nil}, \pi_2] \ Q(\text{val}, \text{wt})$



$$\langle [\text{nil}, \text{cpr } \langle h_1, h_2 \rangle \text{ merge } R \text{ thinlist}(Q)] \rangle \text{ minlist}(R)$$

$\text{listcp} = \text{wrap} + \text{cpr}$ ,  $g_i = [\text{list}(\text{nil}), h_i] - [\text{nil}, \text{cons}] =$   
 $[\text{nil}, \text{cons}], [\text{nil}, \pi_2](l(D)) = [], [\text{nil}, \pi_2](r(p)) = \pi_2(p)$ ,  
 $[\text{nil}, \pi_2] = [\text{nil}, \pi_2]$ ;  $\text{minlist}(R)$  is head, packings coming out in  
 descending value

## §8.5. The paragraph problem

(8.5a)

|    | definition                                                                                                                                                 | type                       | meaning                                                                          |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------|
| 1  | $\text{Line} \triangleq \text{L}(\text{Word})$                                                                                                             | Type                       | a line is a non-empty list of words                                              |
| 2  | $\text{Para} \triangleq \text{L}(\text{Line})$                                                                                                             | Type                       | a paragraph is a non-empty list of lines                                         |
| 3  | $\text{new}(a, \text{xss}) \triangleq [[a]] \# \text{xss}$                                                                                                 | Para                       | start a new first line holding only the word a                                   |
| 4  | $\text{glue}(a, [\text{xs}]) = [[a] \# \text{xs}], \text{glue}(a, [\text{xs}] \# a') = [[a] \# \text{xs}] \# a'$                                           | Para $\rightarrow$ Para    | put the word a at the front of the first line                                    |
| 5  | $[\text{wrap wrap}, \text{new}](\text{l}(a)) = [[a]], [\text{wrap wrap}, \text{new}](\text{r}(q)) = \text{new}(\pi_1(q), \pi_2(q))$                        | F(Para) $\rightarrow$ Para | one word is a one-line paragraph; each further word starts a new line            |
| 6  | $[\text{wrap wrap}, \text{glue}](\text{l}(a)) = [[a]], [\text{wrap wrap}, \text{glue}](\text{r}(q)) = \text{glue}(\pi_1(q), \pi_2(q))$                     | F(Para) $\rightarrow$ Para | one word is a one-line paragraph; each further word joins the first line         |
| 7  | $[\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}] \triangleq [\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}]$       | F(Para) $\rightarrow$ Para | each further word either starts a new line or joins the first line               |
| 8  | $\text{partition} \triangleq ([\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}])$                                                        | L(Word) $\rightarrow$ Para | every way of breaking the words into lines                                       |
| 9  | $\text{width}[a] = \text{len}(a), \text{width}([a] \# \text{xs}) = (\text{len}(a) + \text{width}(\text{xs})) + 1$                                          | Line $\rightarrow$ Int     | the lengths of the words plus one space between neighbours                       |
| 10 | $\text{head}[\text{xs}] = \text{xs}, \text{head}([\text{xs}] \# a) = \text{xs}$                                                                            | Para $\rightarrow$ Line    | the first line of the paragraph                                                  |
| 11 | $\text{fits}(w, [\text{xs}]) = \text{width}(\text{xs}) \leq w, \text{fits}(w, [\text{xs}] \# a) = \text{width}(\text{xs}) \leq w \wedge \text{fits}(w, a)$ | P(Para)                    | every line is at most w wide                                                     |
| 12 | $\text{fits}(w)(\text{xss}, \text{yss}) \iff \text{xss} = \text{yss} \wedge \text{fits}(w, \text{xss})$                                                    | Para $\rightarrow$ Para    | the paragraphs whose every line is at most w wide                                |
| 13 | $\text{ok}(w)(\text{xss}, \text{yss}) \iff \text{xss} = \text{yss} \wedge \text{width}(\text{head}(\text{xss})) \leq w$                                    | Para $\rightarrow$ Para    | the paragraphs whose first line is at most w wide                                |
| 14 | $\text{sqr}(n) \triangleq nn$                                                                                                                              | Int                        | n squared                                                                        |
| 15 | $\text{waste}(w, [\text{xs}]) = 0, \text{waste}(w, [\text{xs}] \# a) = \text{sqr}(w - \text{width}(\text{xs})) + \text{waste}(w, a)$                       | Para $\rightarrow$ Int     | the squared white space left on every line but the last, added up                |
| 16 | $\text{R} = \text{waste } w \leq \text{waste } w^\circ$                                                                                                    | Para $\rightarrow$ Para    | p wastes no more than q                                                          |
| 17 | $\text{Q}(\text{len}, w) = \text{R } n (\text{head } \text{head}^\circ)$                                                                                   | Para $\rightarrow$ Para    | p wastes no more than q and has the same first line                              |
| 18 | $\text{start}(a) \triangleq [[[a]]]$                                                                                                                       | [Para]                     | the one candidate for a single word: the paragraph of one line holding that word |

(8.5b)

| the law                                                                                                                                                                                                                                                                                                                                                    | what it says                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 1 $(1 \times R) \text{ glue} \sqsubseteq \text{glue } R \quad \text{FALSE}$                                                                                                                                                                                                                                                                                | glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this |
| 2 $F(Q(\text{len}, w)) \text{ [wrap wrap, new]} \sqsubseteq \text{[wrap wrap, new]} Q(\text{len}, w)$<br>$(\forall a. 0 \leq \text{len}(a)) \Rightarrow F(Q(\text{len}, w)) \text{ [wrap wrap, glue]} \text{ ok}(w) \sqsubseteq \text{[wrap wrap, glue]} \text{ ok}(w) Q(\text{len}, w)$<br>cons monotonic on $\text{collect} \leq \text{collect}^\circ$ . | both halves are monotonic on Q once ties in waste are broken by the first line                                           |
| 3 $\text{merge}(\tau) = \text{cat}$                                                                                                                                                                                                                                                                                                                        | $\tau$ needs no sorting at all                                                                                           |
| 4 $F(\tau) \text{ [wrap wrap, new]} \sqsubseteq \text{[wrap wrap, new]} \tau$<br>$F(\tau) \text{ [wrap wrap, glue]} \sqsubseteq \text{[wrap wrap, glue]} \tau$                                                                                                                                                                                             | both algebras are monotonic on $\tau$ , so $\tau$ is the sort order P                                                    |

## §8.5.1. Derivation

| $(\forall a. 0 \leq \text{len}(a)) \wedge (\forall a. \text{len}(a) \leq w) \Rightarrow (\text{listcp } (g_1, g_2) \text{ merge}(\tau) \text{ thinlist}(Q(\text{len}, w)))$ (8.5.1a)<br>$\text{minlist}(R) \sqsubseteq \frac{\text{partition fits}(w)}{\exists} \text{ est}(R)$<br>a paragraph laid out as a fold that thins the layouts kept at each word                                                                                                                                                                                                                                                                                                                                                                              |                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>circuit</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Hinze–Marsden</b> |
| $\text{L}(\text{Word}) \xrightarrow{\frac{1}{3}} \text{P}(\text{L}(\text{Word})) \xrightarrow{\text{E}(\text{partition fits}(w))} \text{P}(\text{L}(\text{Line})) \xrightarrow{\text{est}(R)} \text{L}(\text{Line})$ $\frac{\text{partition fits}(w)}{\exists} \text{ est}(R)$                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                      |
| $\text{L}(\text{Word}) \xrightarrow{\frac{1}{3}} \text{P}(\text{L}(\text{Word})) \xrightarrow{\text{E}([\text{wrap wrap, new}] \cup ([\text{wrap wrap, glue}] \text{ok}(w)))} \text{P}(\text{L}(\text{Line})) \xrightarrow{\text{est}(R)} \text{L}(\text{Line})$ $= \frac{([\text{wrap wrap, new}] \cup ([\text{wrap wrap, glue}] \text{ok}(w)))}{\exists} \text{ est}(R)$ fusion, every word fits on a line by itself.                                                                                                                                                                                                                                                                                                                 |                      |
| $\text{L}(\text{Word}) \xrightarrow{\frac{1}{3}} \text{P}(\text{L}(\text{Word})) \xrightarrow{\text{E}([\text{wrap wrap, new}] \cup ([\text{wrap wrap, glue}] \text{ok}(w)))} \text{P}(\text{L}(\text{Line})) \xrightarrow{\text{est}(R)} \text{L}(\text{Line})$ $= \frac{([\text{wrap wrap, new}] \cup ([\text{wrap wrap, glue}] \text{ok}(w)))}{\exists} \text{ est}(R)$ the algebra as $(f_1 p_1) \cup (f_2 p_2)$ , $p_1 \triangleq 1$ — binary thinning theorem.                                                                                                                                                                                                                                                                    |                      |
| $\text{L}(\text{Word}) \xrightarrow{\text{P}(\text{listcp } (g_1, g_2) \text{ merge}(\tau) \text{ thinlist}(Q(\text{len}, w)))} \text{listcp } (g_1, g_2) \text{ merge}(\tau) \text{ thinlist}(Q(\text{len}, w))$ $\text{minlist}(R)$ <p>□ binary thinning theorem, at <math>P \triangleq T</math> with <math>\text{merge } \tau = \text{cat}</math> (<math>\text{merge}(\tau) = \text{cat}</math>), <math>Q</math> from <math>F(Q(\text{len}, w)) \text{ [wrap wrap, new]} \sqsubseteq \text{[wrap wrap, new]} Q(\text{len}, w)</math>, <math>(\forall a. 0 \leq \text{len}(a)) \Rightarrow F(Q(\text{len}, w)) \text{ [wrap wrap, glue]} \text{ ok}(w) \sqsubseteq \text{[wrap wrap, glue]} \text{ ok}(w) Q(\text{len}, w)</math></p> |                      |
| $Q[\text{start}, \text{cpr } (h_1, h_2) \text{ cat thinlist}(Q)] \text{ minlist}(R)$ $\text{listcp} = \text{wrap} + \text{cpr}, g_i \text{ along the coproduct} \text{ — } [\text{wrap wrap, new}]$ $= (\text{l}(a)) = [[a]], [\text{wrap wrap, new}](r(q)) = \text{new}(\pi_1(q), \pi_2(q)), [\text{wrap wrap, glue}](\text{l}(a)) = [[a]], [\text{wrap wrap, glue}](r(q)) = \text{glue}(\pi_1(q), \pi_2(q))$                                                                                                                                                                                                                                                                                                                          |                      |

## §8.6. Bitonic tours

(8.6a)

|    | definition                                                                                                                                                                 | type                                                                  | meaning                                                                             |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1  | $\text{Journey}(\text{City}) \triangleq \mu F$                                                                                                                             | Type                                                                  | a list of at least two cities, the last two held as a pair                          |
| 2  | $\text{Tour}(\text{City}) \triangleq \text{Journey}(\text{City}) \times \text{Journey}(\text{City})$                                                                       | Type                                                                  | the outward journey and the return journey                                          |
| 3  | $\text{head}(\text{wrap}(p)) = \pi_1(p), \text{head}([a]\#a') = a$                                                                                                         | $\text{Journey}(\text{City}) \rightarrow \text{City}$                 | the first city of the journey                                                       |
| 4  | $\text{next}(\text{wrap}(p)) = \pi_2(p), \text{next}([a]\#a') = \text{head}(a')$                                                                                           | $\text{Journey}(\text{City}) \rightarrow \text{City}$                 | the second city of the journey                                                      |
| 5  | $[a]\#\text{tail}(\text{wrap}(p)) = \text{wrap}(a, \pi_2(p)), [a]\#\text{tail}([a']\#a'') = [a]\#a''$                                                                      | $\text{Journey}(\text{City}) \rightarrow \text{Journey}(\text{City})$ | replace the first city of the journey by a                                          |
| 6  | $\text{outcost}(\text{wrap}(p)) = \text{tc}(p), \text{outcost}([a]\#a') = \text{tc}(a, \text{head}(a')) + \text{outcost}(a')$                                              | $\text{Journey}(\text{City}) \rightarrow \text{Int}$                  | the cost of travelling the journey forwards                                         |
| 7  | $\text{incost}(\text{wrap}(p)) = \text{tc}(\pi_2(p), \pi_1(p)), \text{incost}([a]\#a') = \text{tc}(\text{head}(a'), a) + \text{incost}(a')$                                | $\text{Journey}(\text{City}) \rightarrow \text{Int}$                  | the cost of travelling the journey backwards                                        |
| 8  | $\text{cost}(t) \triangleq \text{outcost}(\pi_1(t)) + \text{incost}(\pi_2(t))$                                                                                             | Int                                                                   | the outward journey forwards plus the return journey backwards                      |
| 9  | $\text{start}(a) \triangleq (\text{wrap}(a), \text{wrap}(a))$                                                                                                              | $\text{City} \times \text{City} \rightarrow \text{Tour}(\text{City})$ | the tour of the last two cities, both journeys being that one edge                  |
| 10 | $\text{dropl}(a, t) \triangleq ([a]\#\text{tail}(\pi_1(t)), [a]\#\pi_2(t))$                                                                                                | $\text{Tour}(\text{City})$                                            | a replaces the first city of the outward journey and is put in front of the return  |
| 11 | $\text{dropr}(a, t) \triangleq ([a]\#\pi_1(t), [a]\#\text{tail}(\pi_2(t)))$                                                                                                | $\text{Tour}(\text{City})$                                            | a is put in front of the outward journey and replaces the first city of the return  |
| 12 | $[\text{start}, \text{dropl}](l(p)) = \text{start}(p), [\text{start}, \text{dropl}](r(q)) = \text{dropl}(\pi_1(q), \pi_2(q))$                                              | $F(\text{Tour}(\text{City})) \rightarrow \text{Tour}(\text{City})$    | start at the last two cities, then add each city by dropl                           |
| 13 | $[\text{start}, \text{dropr}](l(p)) = \text{start}(p), [\text{start}, \text{dropr}](r(q)) = \text{dropr}(\pi_1(q), \pi_2(q))$                                              | $F(\text{Tour}(\text{City})) \rightarrow \text{Tour}(\text{City})$    | start at the last two cities, then add each city by dropr                           |
| 14 | $[\text{start}, \text{dropl}] \cup [\text{start}, \text{dropr}] \triangleq [\text{start}, \text{dropl}] \cup [\text{start}, \text{dropr}]$                                 | $F(\text{Tour}(\text{City})) \rightarrow \text{Tour}(\text{City})$    | each further city is added by dropl or by dropr                                     |
| 15 | $\langle [\text{start}, \text{dropl}] \cup [\text{start}, \text{dropr}] \rangle \triangleq \langle [\text{start}, \text{dropl}] \cup [\text{start}, \text{dropr}] \rangle$ | $\mu F \rightarrow \text{Tour}(\text{City})$                          | every bitonic tour of the cities                                                    |
| 16 | $R = \text{cost} \leq \text{cost}^\circ$                                                                                                                                   | $\text{Tour}(\text{City}) \rightarrow \text{Tour}(\text{City})$       | t costs no more than t'                                                             |
| 17 | $\text{next2}(t) \triangleq (\text{next}(\pi_1(t)), \text{next}(\pi_2(t)))$                                                                                                | $\text{City} \times \text{City}$                                      | the second cities of both journeys                                                  |
| 18 | $\text{head2}(t) \triangleq (\text{head}(\pi_1(t)), \text{head}(\pi_2(t)))$                                                                                                | $\text{City} \times \text{City}$                                      | the first cities of both journeys                                                   |
| 19 | $Q = R \cap (\text{next2 next2}^\circ) \cap (\text{head2 head2}^\circ)$                                                                                                    | $\text{Tour}(\text{City}) \rightarrow \text{Tour}(\text{City})$       | t costs no more than t', and both journeys of t and t' share their first two cities |

(8.6b)

| the law                                                                                                                                          | what it says                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1 $(1 \times R) \text{ drop} \sqsubseteq \text{drop} \ R \text{ FALSE}$<br>$(1 \times R) \text{ drop} \sqsubseteq \text{drop} \ R \text{ FALSE}$ | neither drop is monotonic on R: the two edges it adds and removes depend on head and next of both lists            |
| 2 $F(Q) \text{ [start, drop]} \sqsubseteq \text{[start, drop]} \ Q$<br>$F(Q) \text{ [start, drop]} \sqsubseteq \text{[start, drop]} \ Q$ ,       | both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input |
| 3 $F(T) \text{ [start, drop]} \sqsubseteq \text{[start, drop]} \ T$<br>$F(T) \text{ [start, drop]} \sqsubseteq \text{[start, drop]} \ T$         | both algebras are monotonic on T, so T is the sort order P                                                         |

## §8.6.1. Derivation

| $(\text{listcp } \langle g_1, g_2 \rangle \text{ cat thinlist}(Q)) \text{ minlist}(R) \sqsubseteq \frac{(\text{[start, drop]} \sqcup \text{[start, drop]})}{\exists} \text{ est}(R) \quad (8.6.1a)$ <p>a least-cost bitonic tour, as a fold that thins the tours kept at each city</p>                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Hinze–Marsden                                                                                                                                                       |
| $\mu F \left[ \frac{1}{\exists} \right] \text{P}(\mu F) \left[ \frac{E(\text{[start, drop]} \sqcup \text{[start, drop]})}{\exists} \right] \text{P}(\text{Journey}(\text{City})^2) \text{est}(R) \text{Journey}(\text{City})$ $\frac{(\text{[start, drop]} \sqcup \text{[start, drop]})}{\exists} \text{ est}(R)$                                                                                                                                                                                                                                                                                                                       | <p>Journey City</p> <p><math>\nabla (\text{[start, drop]} \sqcup \text{[start, drop]})</math></p> <p>est(R)</p> <p>Journey×Journey City</p>                         |
| $\mu F \left[ \frac{1}{\exists} \right] \text{P}(\mu F) \left[ \frac{E(\text{[start, drop]} \sqcup \text{[start, drop]})}{\exists} \right] \text{P}(\text{Journey}(\text{City})^2) \text{est}(R) \text{Journey}(\text{City})$ $= \frac{(\text{[start, drop]} \sqcup \text{[start, drop]})}{\exists} \text{ est}(R)$ $= (\text{[start, drop]} \sqcup \text{[start, drop]}) \triangleq (\text{[start, drop]} \sqcup \text{[start, drop]})$                                                                                                                                                                                                |                                                                                                                                                                     |
| $\mu F \left[ \frac{F(\text{list}(\text{Journey}(\text{City})^2)) \text{listcp} \text{list}(F(\text{Journey}(\text{City})^2))}{\exists} \right] \text{cat} \text{thinlist}(Q) \text{list}(\text{Journey}(\text{City})^2) \text{minlist}(R) \text{Journey}(\text{City})$ $(\text{listcp } \langle g_1, g_2 \rangle \text{ cat thinlist}(Q)) \text{ minlist}(R)$ <p>binary thinning theorem, at <math>P \triangle T</math> with merge <math>T = \text{cat}</math>, <math>Q</math> from <math>F(Q)</math></p> $\text{[start, drop]} \sqsubseteq \text{[start, drop]} \ Q, F(Q) \text{ [start, drop]} \sqsubseteq \text{[start, drop]} \ Q$ | <p>Journey City</p> <p>list</p> <p><math>(\text{listcp } \langle g_1, g_2 \rangle \text{ cat thinlist}(Q))</math></p> <p>minlist(R)</p> <p>Journey×Journey City</p> |
| $(\text{[start wrap, cpr } (\text{list}(\text{drop}_1), \text{list}(\text{drop}_2)) \text{ cat thinlist}(Q)) \text{ minlist}(R)$ $\text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{start}), \text{list}(\text{drop}_i)] \text{ —}$ $= \text{[start, drop]}(\text{l}(p)) = \text{start}(p), \text{[start, drop]}(\text{r}(q)) = \text{drop}_1(\pi_1(q), \pi_2(q)), \text{[start, drop]}(\text{l}(p)) = \text{start}(p),$ $\text{[start, drop]}(\text{r}(q)) = \text{drop}_2(\pi_1(q), \pi_2(q)); \text{quadratic, two tours added per step}$                                                                           |                                                                                                                                                                     |